APL wiring per glomerulus: the connectome numbers the model uses

> **QUESTION**
> How much APL wiring does each antennal-lobe glomerulus get in the calyx, measured per unit of its input to the KCs, and is that pattern the same in hemibrain and male CNS?

INPUTS

- All in `with_outside_ROI_synapses/claws_v2m2/` (corrected calyx, new claws; Report 02):
- `PN_calyx_weights_APL_vs_KC.csv`: per PN, its calyx synapses onto KCs and to and from APL (Report 01)
- `odor_weighted/KC_APL_synapses_on_claws_by_glomerulus.csv`: KC↔APL synapses that sit on a claw, with the claw's glomerulus (Report 02)
- `hemibrain_KC_claws_derived.csv.gz`, `malecns_KC_claws_derived.csv.gz`: the claws (Report 02)
- `KC_weights_APL_PN_per_KC.csv`: per KC, its calyx APL synapses (Report 01)
- For Section 5: the corrected-calyx PN ↔ KC and PN ↔ APL synapses of all four connectomes

OUTPUTS

- `with_outside_ROI_synapses/claws_v2m2/odor_weighted/APL_wiring_by_glomerulus.csv` (one row per dataset × glomerulus)
- the weights p_g, q_g and the bouton share f used in Report 05
- `reports/data/r04v2_pg_per_calyx.csv`, `r04v2_PN_calyx_weights_per_side.csv` (p_g per calyx, all connectomes)

SCRIPTS

- `scripts/apl_block_model.py`, section "1. APL wiring per glomerulus" (`APL_VARIANT=claws_v2m2`)
- `scripts/reports/r04_prep_v2.py` (per-calyx counts)
- `scripts/reports/r04_glomerulus_wiring.py` (this report)

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Report 05 lets APL's inhibition differ between glomeruli in three of its versions (P, L and M). This report explains where those per-glomerulus numbers come from. The model uses only hemibrain and male CNS (Yang, 2026-10-21), with the corrected calyx and the new claws (Report 02). Section 5 adds the bouton weight for every calyx of all four connectomes, as Yang asked. In the male CNS the two calyces are pooled here, because the per-glomerulus numbers are computed per dataset.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range
g	An antennal-lobe glomerulus. Only uniglomerular PN types count (one glomerulus in the type name).	hemibrain names VC3l, VC3m, VC5 are renamed VC3, VC5, VM6 to match the other datasets
$\mathrm{PN}(g)$	The uniglomerular PNs of glomerulus g	
$\mathrm{claws}(g)$	Claws whose main PN belongs to g (Report 02)	

Symbol	What it stands for	Values, units, range
$n_g^{\mathrm{PN}\to\mathrm{KC}}$	PN→KC synapses of $\mathrm{PN}(g)$ in the calyx (onto any KC)	synapse count
n_g^{claw}	PN synapses on $\mathrm{claws}(g)$ (from any PN)	synapse count
$\lvert\mathrm{claws}(g)\rvert$	Number of claws of g	count
p_g	Bouton weight: APL→PN synapses on $\mathrm{PN}(g)$ in the calyx ÷ $n_g^{\mathrm{PN}\to\mathrm{KC}}$	synapses per synapse
q_g	Claw weight: APL→KC synapses on $\mathrm{claws}(g)$ ÷ n_g^{claw}	synapses per synapse
f	Boutons' share of APL's calyx output: APL→PN ÷ (APL→PN + APL→KC), all calyx synapses	per dataset
ρ_s	Spearman rank correlation, here between datasets over glomeruli	−1 to 1

1 The units: PN boutons, claws and microglomeruli

In the calyx, each PN axon forms **boutons**: swellings that carry its output synapses. Several KC dendritic **claws** wrap around one bouton, and the bouton with its claws is called a **microglomerulus**. APL makes synapses in both directions with both partners: APL→PN and PN→APL on the boutons, and APL→KC and KC→APL on the claws (Figure 1). Report 02 explains how they are found. Boutons are clusters of a PN's calyx T-bars within 2 µm. Claws follow Yang's corrected rule: a KC's calyx synapse sites are clustered at 8 µm along the KC skeleton, and each cluster is split by presynaptic PN (each PN with ≥ 2 PSDs gets its own claw). A KC↔APL synapse counts as **on a claw** if it lies within 8 µm, along the KC, of a PN synapse of that claw. Every claw belongs to one glomerulus: that of its PN.

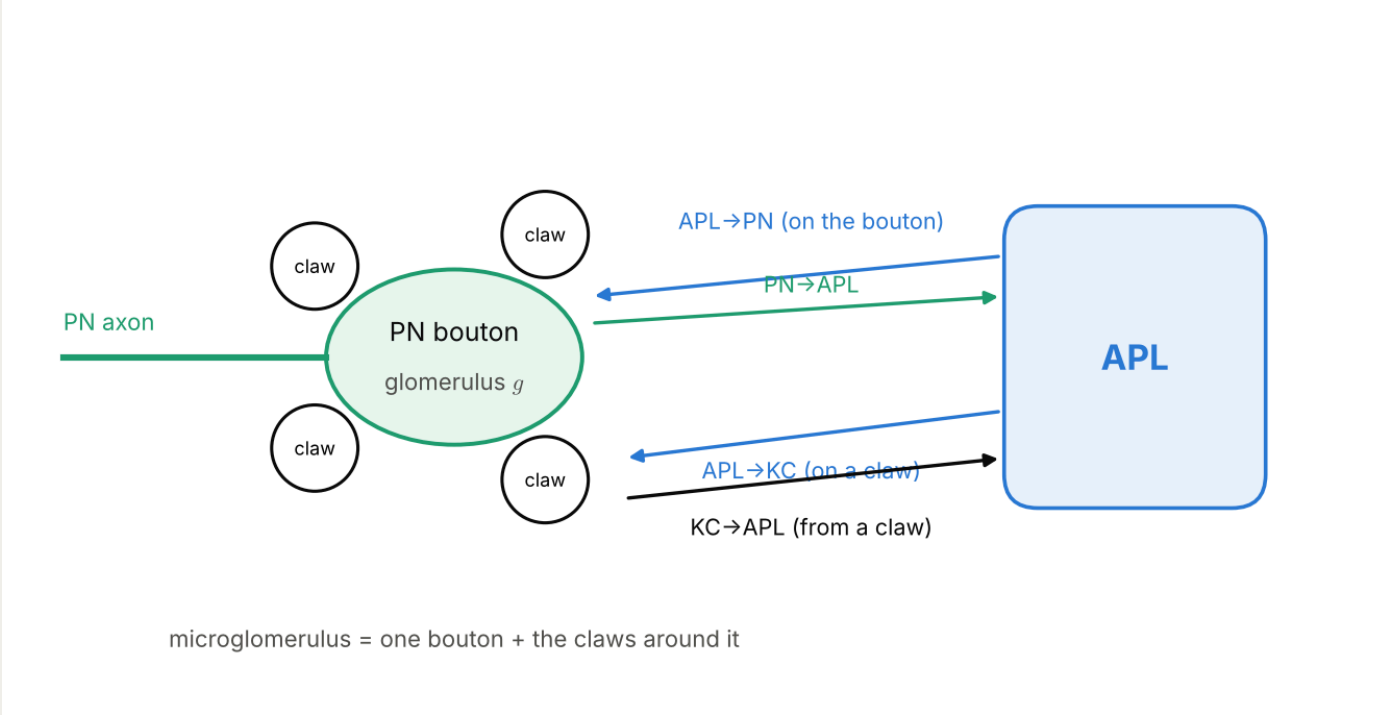


Figure 1. The four kinds of APL synapse at a microglomerulus. One PN bouton (green, belonging to glomerulus g) with KC claws around it, forming a microglomerulus. APL (blue) makes synapses in four directions there. APL→PN and PN→APL are on the bouton; APL→KC and KC→APL are on the claws. Arrows point from the presynaptic to the postsynaptic cell. The drawing is schematic and not to scale.

2 Six per-glomerulus numbers

Raw synapse counts mostly reflect how big a glomerulus is: how many PNs it has and how many claws. To compare glomeruli, each count is divided by a measure of the glomerulus's input to the KCs. For glomerulus g:

$$p_g = \frac{\#\,\text{APL}\to\text{PN synapses on PN}(g)}{n_g^{\text{PN}\to\text{KC}}}, \qquad p_g^{\text{in}} = \frac{\#\,\text{PN}\to\text{APL synapses from PN}(g)}{n_g^{\text{PN}\to\text{KC}}} \tag{1}$$

all synapses in the calyx; $n_g^{\text{PN}\to\text{KC}}$ = PN→KC synapses of the same PNs

$$q_g = \frac{\#\,\text{APL}\to\text{KC on claws}(g)}{n_g^{\text{claw}}}, \qquad q_g^{\text{in}} = \frac{\#\,\text{KC}\to\text{APL on claws}(g)}{n_g^{\text{claw}}} \tag{2}$$

n_g^{claw} = PN synapses on those claws

Two more divide the claw counts by the number of claws instead (c_g^{out} and c_g^{in}, APL→KC and KC→APL synapses per claw). The model uses only p_g (version P, and M) and q_g (version L). p_g measures how much APL inhibition a glomerulus's boutons receive for each synapse they pass on to KCs. q_g measures how much APL inhibition its claws receive for each PN synapse they get.

Table 2. Totals over uniglomerular glomeruli. Calyx synapses only; male CNS includes both calyces.

dataset	glomeruli	PNs	PN→KC synapses	APL→PN	PN→APL	claws	claw PN synapses	APL→KC on claws	KC→APL on claws
hemibrain	57	117	178,098	2,669	2,677	10,766	176,656	16,957	10,239
male CNS	57	243	377,144	4,217	5,068	21,245	375,795	24,664	20,445

Small glomeruli give noisy ratios, so the comparisons below use the 51 glomeruli with at least 40 claws and at least 500 PN→KC synapses in both datasets. The model itself uses every glomerulus that has a value.

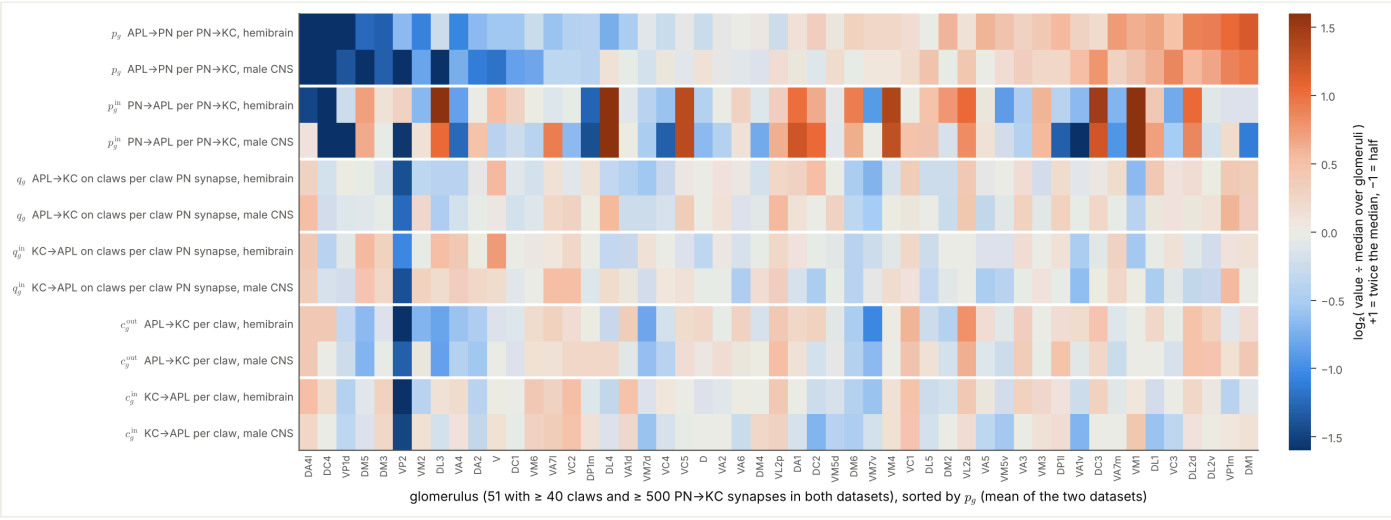


Figure 2. APL wiring per glomerulus, relative to the median. Each row is one of the six numbers in one dataset, and each column a glomerulus. Color is the value divided by that row's median over glomeruli, on a $\log_2$ scale: orange above the median, blue below, grey-white at the median. The scale is clipped at ±1.6, about 3× above or below, and a value of 0 (hemibrain DC4 has no APL→PN synapse) takes the darkest blue. Columns are sorted by the bouton weight p_g. The top two rows (p_g) show the clearest glomerulus-to-glomerulus differences, in the same order in both datasets.

3 Which numbers vary, and are they conserved?

Table 3. Spread across glomeruli and agreement between datasets (51 glomeruli). The spread is the 10th and 90th percentile over glomeruli divided by the median: "0.5–1.6" means the middle 80% of glomeruli lie between half and 1.6 times the median. The last column is the rank correlation of the glomerulus values between the two datasets.

number	symbol	hemibrain: 10th–90th percentile (× median)	male CNS: 10th–90th percentile (× median)	Spearman ρ_s, hemibrain vs male CNS
APL→PN per PN→KC	p_g	0.47–1.62	0.39–1.54	0.82
PN→APL per PN→KC	p_g^{in}	0.55–2.52	0.40–2.32	0.75
APL→KC on claws per claw PN synapse	q_g	0.75–1.29	0.81–1.27	0.68
KC→APL on claws per claw PN synapse	q_g^{in}	0.78–1.25	0.72–1.26	0.57
APL→KC per claw	c_g^{out}	0.63–1.32	0.68–1.36	0.71
KC→APL per claw	c_g^{in}	0.76–1.33	0.69–1.20	0.70

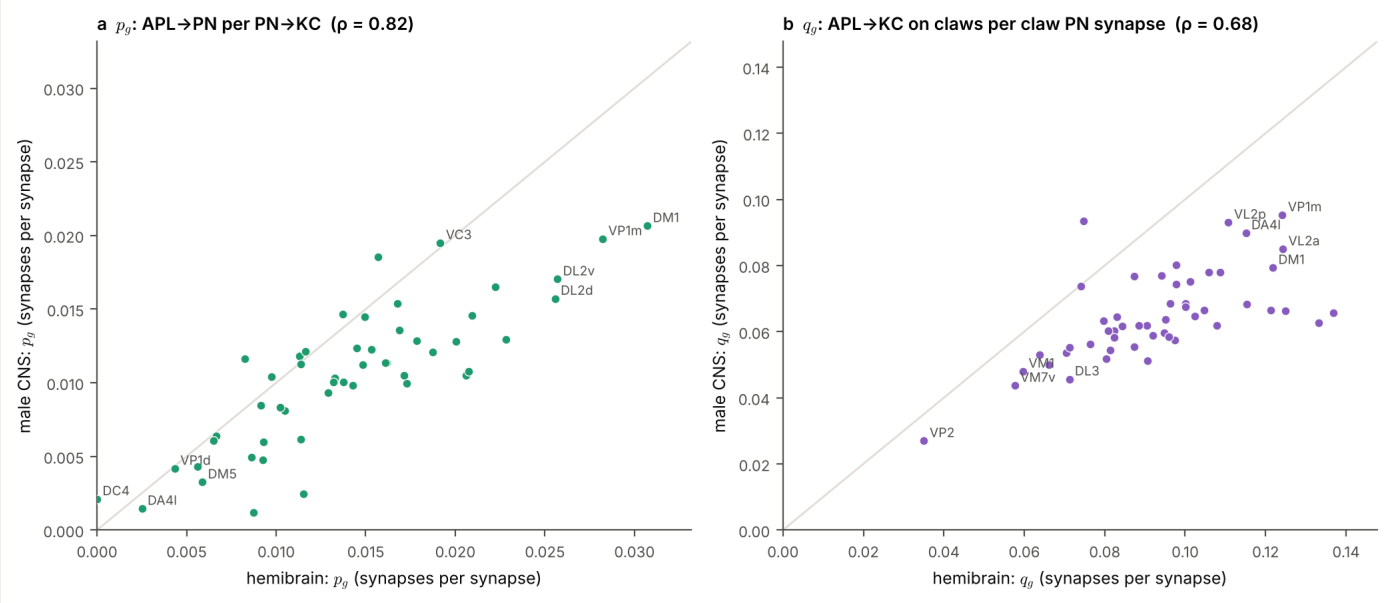


Figure 3. The two weights the model uses, hemibrain against male CNS. Each point is one of the 51 glomeruli. Its value in hemibrain is on the horizontal axis, in male CNS on the vertical axis, and the grey line marks equal values. **a**, Bouton weight p_g: APL→PN synapses per PN→KC synapse. **b**, Claw weight q_g: APL→KC synapses on the claws per claw PN synapse. The labelled glomeruli are the four lowest and five highest (mean of the two datasets, each relative to its median). ρ is the Spearman correlation over glomeruli.

- **Bouton weight** p_g varies 3.4–3.9-fold across glomeruli (10th to 90th percentile; hemibrain, male CNS), in a similar order in both datasets (ρ_s = 0.82). Lowest: DC4, DA4l, VP1d, DM5, VP2. Highest: VC3, DL2d, DL2v, VP1m, DM1.
- **Claw weight** q_g varies only 1.7–1.6-fold (ρ_s = 0.68).
- The other four numbers agree less between the datasets (ρ_s 0.57–0.75). KC→APL on claws per claw PN synapse agrees least.

4 How much of APL's calyx output goes to boutons

Versions M0 and M of Report 05 put only part of the inhibition on the boutons, the same share as APL's synapses:

$$f = \frac{\sum_{\text{PNs}} \#\,\text{APL}{\to}\text{PN}}{\sum_{\text{PNs}} \#\,\text{APL}{\to}\text{PN} + \sum_{\text{KCs}} \#\,\text{APL}{\to}\text{KC}} \tag{3}$$

all calyx synapses of the dataset, not only those on claws or uniglomerular PNs

Table 4. The bouton share f. About one in ten of APL's calyx output synapses is onto a PN; the rest are onto KCs.

dataset	APL→PN in the calyx	APL→KC in the calyx	f
hemibrain	2,902	26,233	0.100
male CNS	4,507	39,247	0.103

5 The bouton weight in every connectome, per calyx

Yang asked for p_g in each connectome, one row per calyx. p_g needs no claws, so it can be computed for all four connectomes. Each calyx is counted on its own: an APL↔PN synapse belongs to the calyx of its APL, and a PN→KC synapse to the calyx of the nearest APL calyx synapse (each APL covers only its own calyx). The counts of the two calyces add up exactly to the per-PN table `PN_calyx_weights_APL_vs_KC.csv` (corrected calyx). A few PN types reach both calyces (V_ilPN, VL1_ilPN, VP1d_il2PN and some multiglomerular types); their synapses are split between the calyces accordingly.

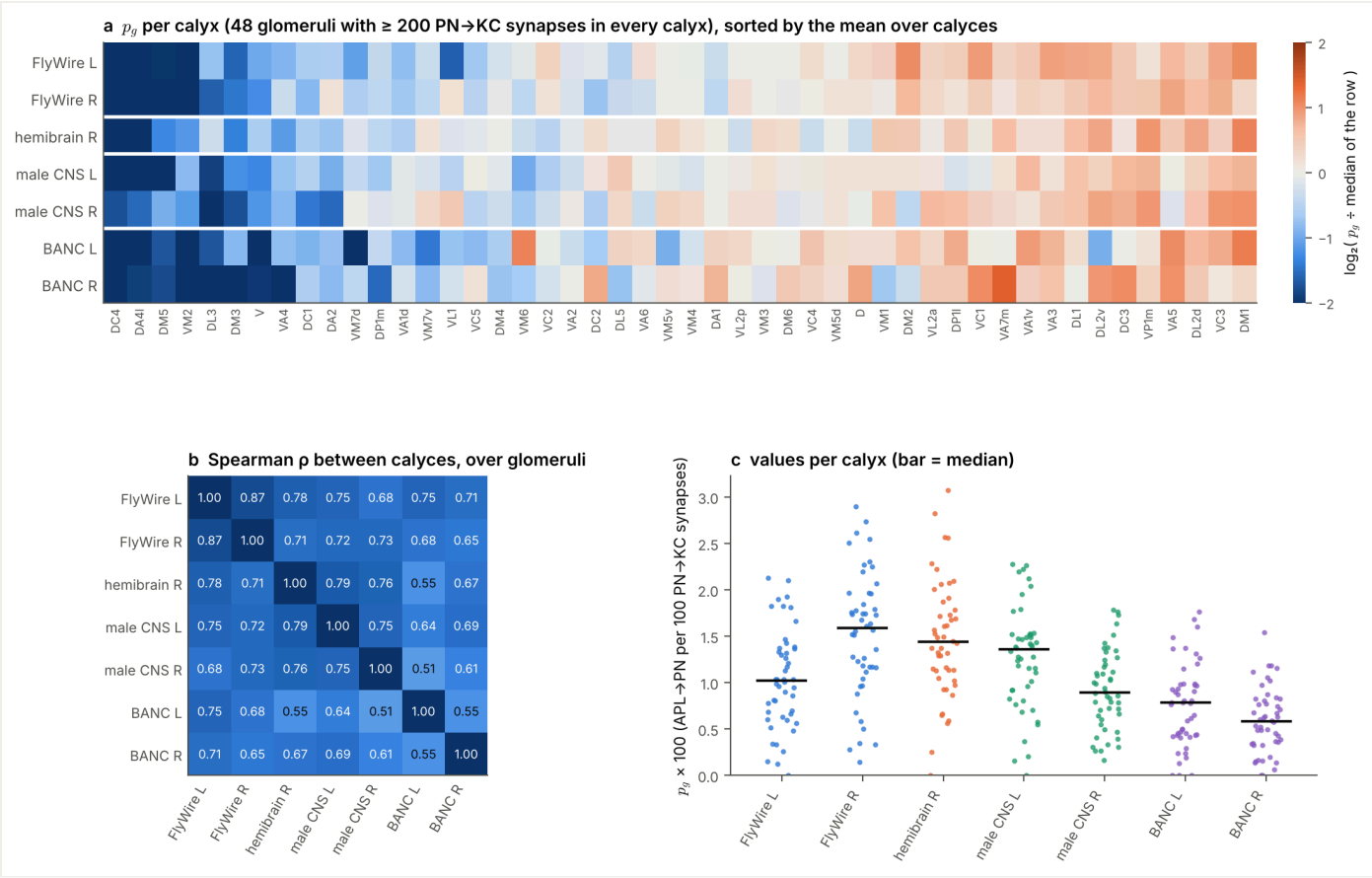


Figure 4. p_g per calyx in every connectome. a, Each row is one calyx and each column one glomerulus. Color is p_g (APL→PN synapses on the glomerulus's PNs in that calyx, per PN→KC synapse of those PNs there) divided by the row's median, on a $\log_2$ scale clipped at ±2 (4× above or below); a value of 0 takes the darkest blue. White lines separate the connectomes. **b**, Rank correlation of the rows of a over the same glomeruli. **c**, The raw values: each dot is one glomerulus; the dot color marks the connectome.

Table 5. The bouton weight per calyx. Uniglomerular PNs only, corrected calyx. Columns 2–5 use every glomerulus of the calyx; the last two use the glomeruli of Figure 4.

Calyx	glomeruli	PN→KC	APL→PN	all glomeruli: APL→PN per 100 PN→KC	median p_g × 100 (48 glomeruli)	10th–90th percentile (× median)
FlyWire L	57	196,590	2,170	1.10	1.02	0.33–1.78
FlyWire R	57	166,295	2,600	1.56	1.59	0.35–1.49
hemibrain R	57	178,098	2,669	1.50	1.44	0.46–1.56
male CNS L	57	185,502	2,429	1.31	1.36	0.42–1.52
male CNS R	57	191,642	1,788	0.93	0.89	0.36–1.62
BANC L	57	79,200	637	0.80	0.78	0.29–1.75
BANC R	56	84,690	495	0.58	0.58	0.25–1.85

- **The pattern is shared across connectomes.** Over the 48 glomeruli the calyces correlate at ρ_s 0.51–0.79 between connectomes, and 0.87, 0.75, 0.55 between the two calyces of FlyWire, male CNS and BANC.

- **The level differs between calyces**, even within one brain: APL→PN per 100 PN→KC synapses ranges 0.58–1.56 (Table 5, all glomeruli), and BANC has the fewest. The pattern across glomeruli, not the level, is what the model uses (p_g enters relative to its mean).
- The glomeruli at the two ends are mostly the same as in Section 3 (Figure 3).

Caveats

Synapse counts stand in for synaptic strength. Nothing here measures how strong a single APL→PN or APL→KC synapse is.

The model numbers (Sections 2–4) pool the two male CNS calyces, because the per-glomerulus weights are computed per dataset.

Glomeruli missing from the response tables still have weights, but they carry no odor input in the model.

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